LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

Unit Test: Sound

Chapter / Topic: Sound (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 45 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks.
3. Frequency = number of oscillations ÷ time · time period = 1 ÷ frequency

SECTION A

6 marks

Choose the correct option.

Q1. Sound is produced by: **[1]**

(a) objects at rest (b) vibrating objects
(c) light (d) heat

Q2. The unit of frequency is the: **[1]**

(a) metre (b) second
(c) kilogram (d) hertz

Q3. The loudness of a sound depends on its: **[1]**

(a) amplitude (b) frequency
(c) time period (d) speed

Q4. Sound cannot travel through: **[1]**

(a) a vacuum (b) air
(c) water (d) steel

Q5. In humans, sound is produced by the: **[1]**

(a) eardrum (b) lungs
(c) nose (d) larynx (voice box)

Q6. A vibration has a time period of 0.05 s. Its frequency is: **[1]**

(a) 0.05 Hz (b) 5 Hz
(c) 20 Hz (d) 50 Hz

SECTION B

8 marks

Q7. Why can't astronauts on the Moon talk to each other directly, without radios? **[2]**

Q8. Which part of the ear vibrates when sound reaches it? Why should we never put sharp objects into our ears? **[2]**

Q9. What is the difference between music and noise? **[2]**

Q10. Give two sources of noise pollution and two of its harmful effects. **[2]**

SECTION C

6 marks

Q11. A pendulum makes 20 oscillations in 4 seconds. **[3]**

(i) Find its frequency. **[1]**

(ii) Find its time period. **[1]**

(iii) If a vibrating object vibrates with a larger amplitude, what changes about the sound it makes? **[1]**

Q12. Which part vibrates to produce sound in (a) a sitar, (b) a flute, (c) a dholak? **[3]**

SECTION D

5 marks

Q13. Answer the following: **[5]**

(i) How is the human voice produced? **[2]**

(ii) Why are men's voices usually deeper than women's? **[1]**

(iii) Suggest two ways to reduce noise near a school. **[2]**

~ All the best! ~